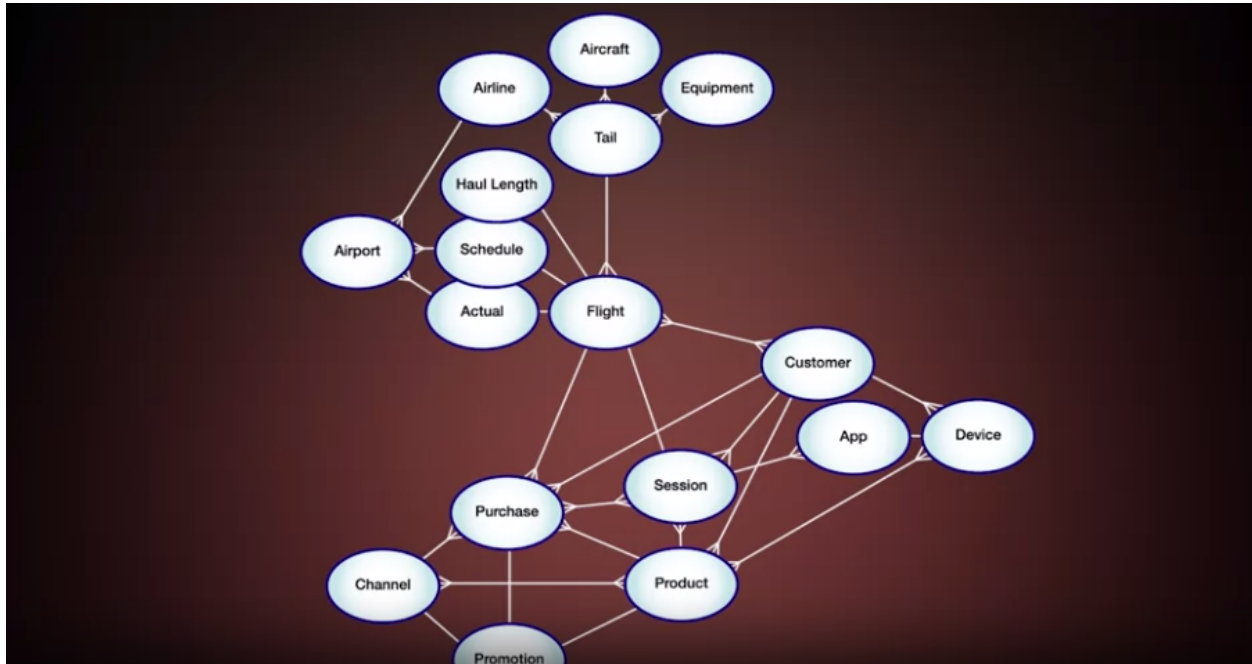


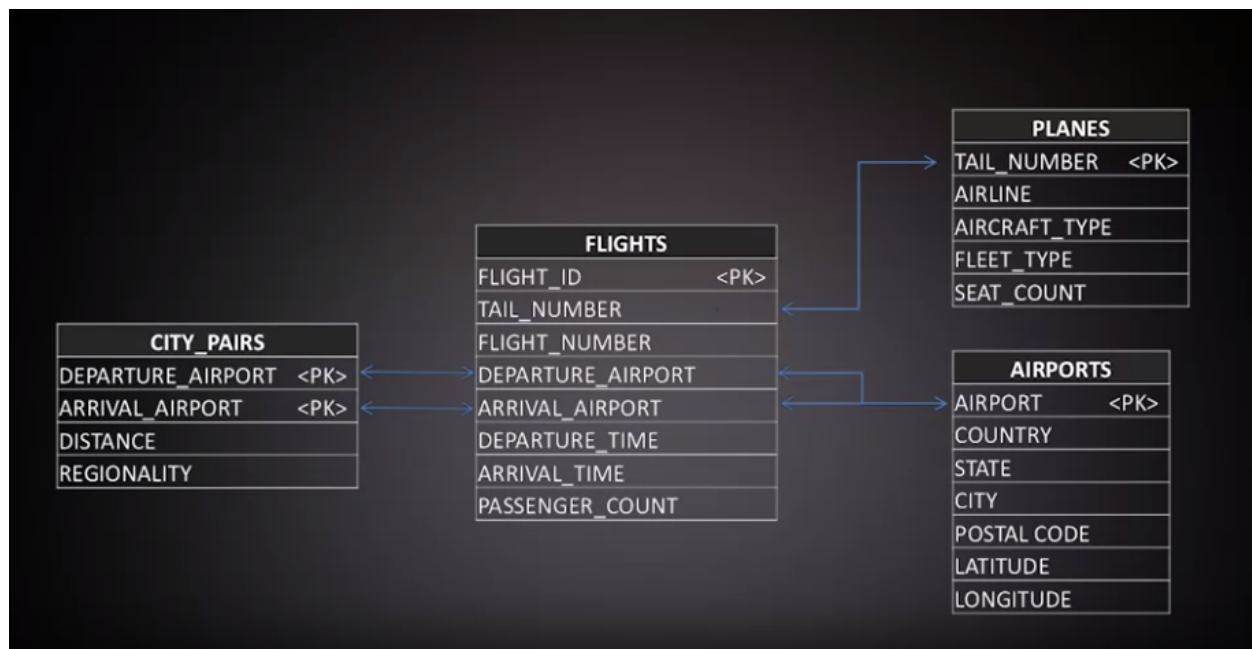
Question: Part 1:

Conceptual business model. Construct a conceptual business model for a specific business that you are familiar with or have interest in. Your model should represent at least 10 ideas. It should visually represent one to one, many to one, or many to many relationships among ideas. It should be like this.



Part 2: Relational data model.

Take a subset of the ideas from the conceptual model you constructed in Part 1 and design a simple relationship model similar to. Your model should have at least 5 tables. You should include at least 20 attributes, or fields, in your model (20 total across all tables, not per table). Your model should be normalized. Identify the primary key in each table, and state whether it is a natural or surrogate key. For each relationship between tables, identify any foreign keys needed to define the relationship. For each table, identify what type of system or systems you think the data might come from, like those we discussed in Module 1, Video 6. It should be like this.



Part 3: SQL queries.

Using the data model you constructed in Part 2, come up with two data extracts you think would be interesting, then write SQL queries to provide each one. For each query, state what data you are trying to get and why it would be interesting. Provide the SQL query using commands and syntax. For maximum credit, at least one of your queries should involve a join across two or more tables.

Part 4: Sensitive data and data quality issues.

Consider the data privacy and data quality implications of the data model you constructed in Part 2. Identify any fields you think might be PII, CFI, CPNI, or PHI. What data elements in your model will present the most significant data quality challenges? Explain your reasoning.